

Testing of Hypothesis

Population:

Every statistical investigation aims at collecting information about some aggregate or collection of individuals or of their attributes, rather than the individual themselves. In statistical language such a collection is called a population.

Example:

- 1) The numbers on cards in a deck. (finite population)
- 2) The total number of blood type of students in a school . (finite population)
- 3) All measurements of the depth of a lake, from any conceivable position (infinite population)

Sample and Sampling:

A sample is a subset of a population and the process of selection of such samples is called sampling.

Example:

- 1) In an organization of 500 employees, if the HR team decides on conducting team building activities, it is highly likely that they would prefer picking chits out of a bowl. In this case, each of the 500 employees has an equal opportunity of being selected.
- 2) a researcher intends to collect a systematic sample of 500 people in a population of 5000. He/she numbers each element of the population from 1-5000 and will choose every 10th individual to be a part of the sample.

Hypothesis Testing

Hypothesis testing or significance testing is a method for testing a claim or hypothesis about a parameter in a population, using data measured in a sample. In this method, we test some hypothesis by determining the likelihood that a sample statistic could have been selected, if the hypothesis regarding the population parameter were true.

Example:

1) The CEO of a large electric utility claims that at least 80 percent of his 10,000,000 customers are very satisfied with the service they receive. To test this claim, the local newspaper surveyed 100 customers, using simple random sampling. Among the sampled customers, 73 percent say they are very satisfied. Based on these findings, can we reject the CEO's hypothesis that 80 percent of the customers are very satisfied?

Example 2

In a large city A, 20 percent of a random sample of 900 school boys had a slight physical defect. In another large city B, 18.5 percent of a random sample of 1600 school boys had the same defect. Is the difference between the proportions significant?

Parameter and statistic

The statistical constants (like mean (μ), variance (σ^2), etc.,) of the population is called as the parameters.

The statistical measures (like mean ($\bar{x}$), variance ($\bar{s}^2$), etc.,) computed from the sample is called as the statistics.

Steps involved in hypothesis testing

Step 1:

we identify a hypothesis or claim that we feel should be tested. In this step we form the null and alternate hypothesis.

Example:

we might want to test the claim that the mean number of hours that children in the United States watch TV is 3 hours.

Null hypothesis

The null hypothesis, symbolized by H_0 , is a statistical hypothesis that states that there is no difference between a parameter and a statistic value or that there is no difference between two parameters.

Example:

For the children watching TV example, we state the null hypothesis that children in the United States watch an average of 3 hours of TV per week.

Alternative Hypothesis

The alternative hypothesis, symbolized by H_1 , is a statistical hypothesis that states a specific difference between a parameter and a specific value or states that there is a difference between two parameters.

In other words, we can say H_1 is complementary to H_0 .

Example:

For the children watching TV example, we may have reason to believe that children watch more than ($>$) or less than ($<$) 3 hours of TV per week. When we are uncertain of the direction, we can state that the value in the null hypothesis is not equal to ($\neq$) 3 hours.

Types of Tests

Two-Tailed Test:

- A medical researcher is interested in finding out whether a new medication will have any undesirable side effects. The researcher is particularly concerned with the pulse rate of the patients who take the medication.
- What are the hypotheses to test whether the pulse rate will be different from the mean pulse rate of 82 beats per minute?
- $H_0 : \mu = 82$ and $H_1 : \mu \neq 82$.

Right-Tailed Test:

- A chemist invents an additive to increase the life of an automobile battery. If the mean lifetime of the battery is 36 months, then his hypotheses are
- $H_0 : \mu = 36$ and $H_1 : \mu > 36$
- This is a **Right-Tailed** test.

Left-Tailed Test:

- A contractor wishes to lower heating bills by using a special type of insulation in houses. If the average of the monthly heating bills is Rs.78, her hypotheses about heating costs will be
- $H_0 : \mu = Rs.78$ and $H_1 : \mu < Rs.78$
- This is a **Left-Tailed** test.

Step 2:

Set the criteria for a decision. To set the criteria for a decision, we state the **level of significance** for a test.

This is similar to the criterion that jurors use in a criminal trial. Jurors decide whether the evidence presented shows guilt *beyond a reasonable doubt* (this is the criterion).

Likewise, in hypothesis testing, we collect data to show that the null hypothesis is not true, based on the likelihood of selecting a sample mean from a population (the likelihood is the criterion).

Level of significance

The likelihood or level of significance is typically set at 5% in behavioral research studies. When the probability of obtaining a sample mean is less than 5% if the null hypothesis were true, then we conclude that the sample we selected is too unlikely and so we reject the null hypothesis.

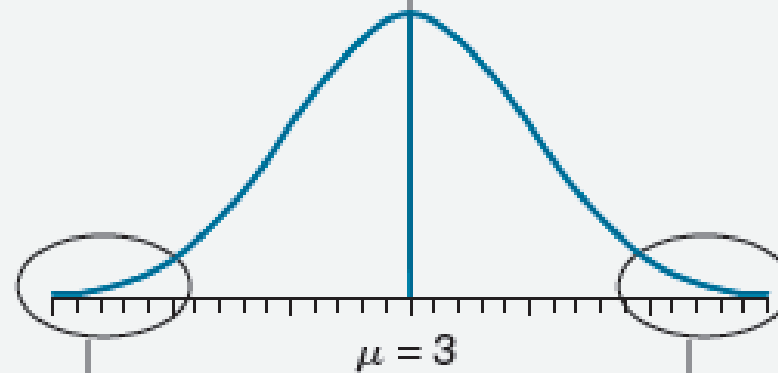
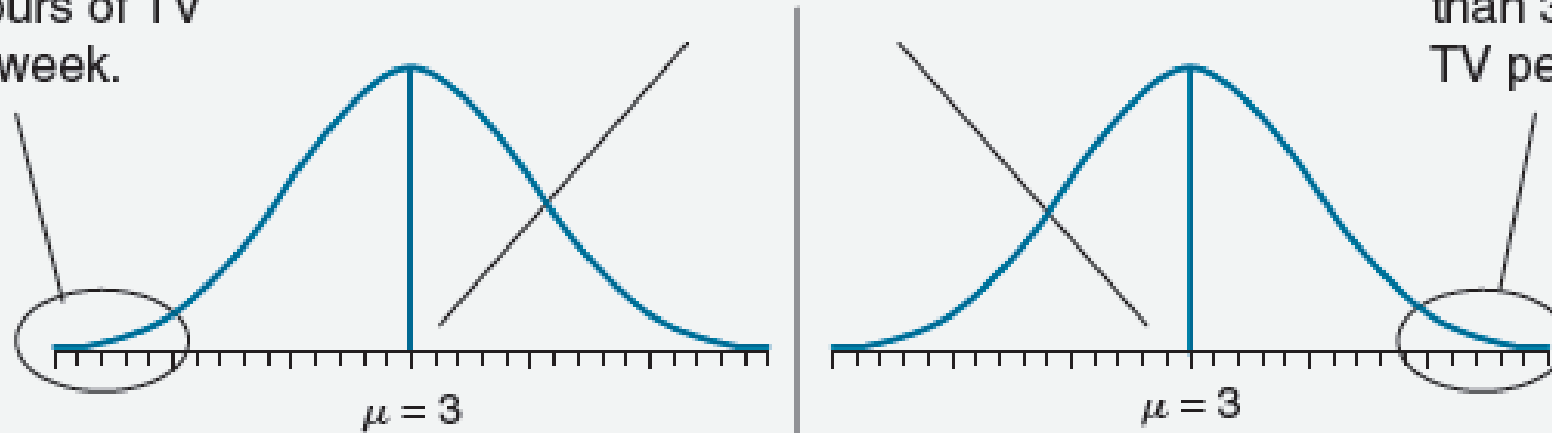
Example:

The claim is that children watch 3 hours of TV per week. Most samples we select should have a mean close to or equal to 3 hours if the claim we are testing is true. So at what point do we decide that the discrepancy between the sample mean and 3 is so big that the claim we are testing is likely not true?

H_1 : Children watch less than 3 hours of TV per week.

We expect the sample mean to be equal to the population mean.

H_1 : Children watch more than 3 hours of TV per week.



H_1 : Children do not watch 3 hours of TV per week.

Step 3:

The **test statistic** is a mathematical formula that allows researchers to determine the likelihood of obtaining sample outcomes if the null hypothesis were true. The value of the test statistic is used to make a decision regarding the null hypothesis.

The numerical value obtained from a test statistic is called the **calculated value**.

Test statistic

A test statistic tells us how far, or how many standard deviations, a sample mean is from the population mean. The larger the value of the test statistic, the further the distance, or number of standard deviations, a sample mean is from the population mean stated in the null hypothesis.

Step 4: Decision Making

We use the value of the test statistic to make a decision about the null hypothesis.

The decision is based on the probability of obtaining a sample mean, given that the value stated in the null hypothesis is true.

Errors in Hypothesis Testing:

- A **Type I error** occurs if one rejects the null hypothesis when it is true. This is similar to a good product being rejected by the consumer and hence Type I error is also known as *producer's risk*.
- The probability of committing Type I error is the **level of significance** (LOS) which is denoted by α . Typical significance levels are: 0.10, 0.05, 0.02 and 0.01.
- A **Type II error** occurs if one does not reject the null hypothesis when it is false. As this error is similar to that of accepting a product of inferior quality, it is known as *consumer's risk*. The probability of committing Type II error is denoted by β .

The critical values for some standard LOS's are given in the following table:

Table 1:

Type of Test	$\alpha = 1\%(0.01)$	$\alpha = 2\%(0.02)$	$\alpha = 5\%(0.05)$	$\alpha = 10\%(0.1)$
Two-Tailed	$ z_\alpha = 2.58$	$ z_\alpha = 2.33$	$ z_\alpha = 1.96$	$ z_\alpha = 1.645$
Right-Tailed	$z_\alpha = 2.33$	$z_\alpha = 2.055$	$z_\alpha = 1.645$	$z_\alpha = 1.28$
Left-Tailed	$z_\alpha = -2.33$	$z_\alpha = -2.055$	$z_\alpha = -1.645$	$z_\alpha = -1.28$

Critical/Rejection Region:

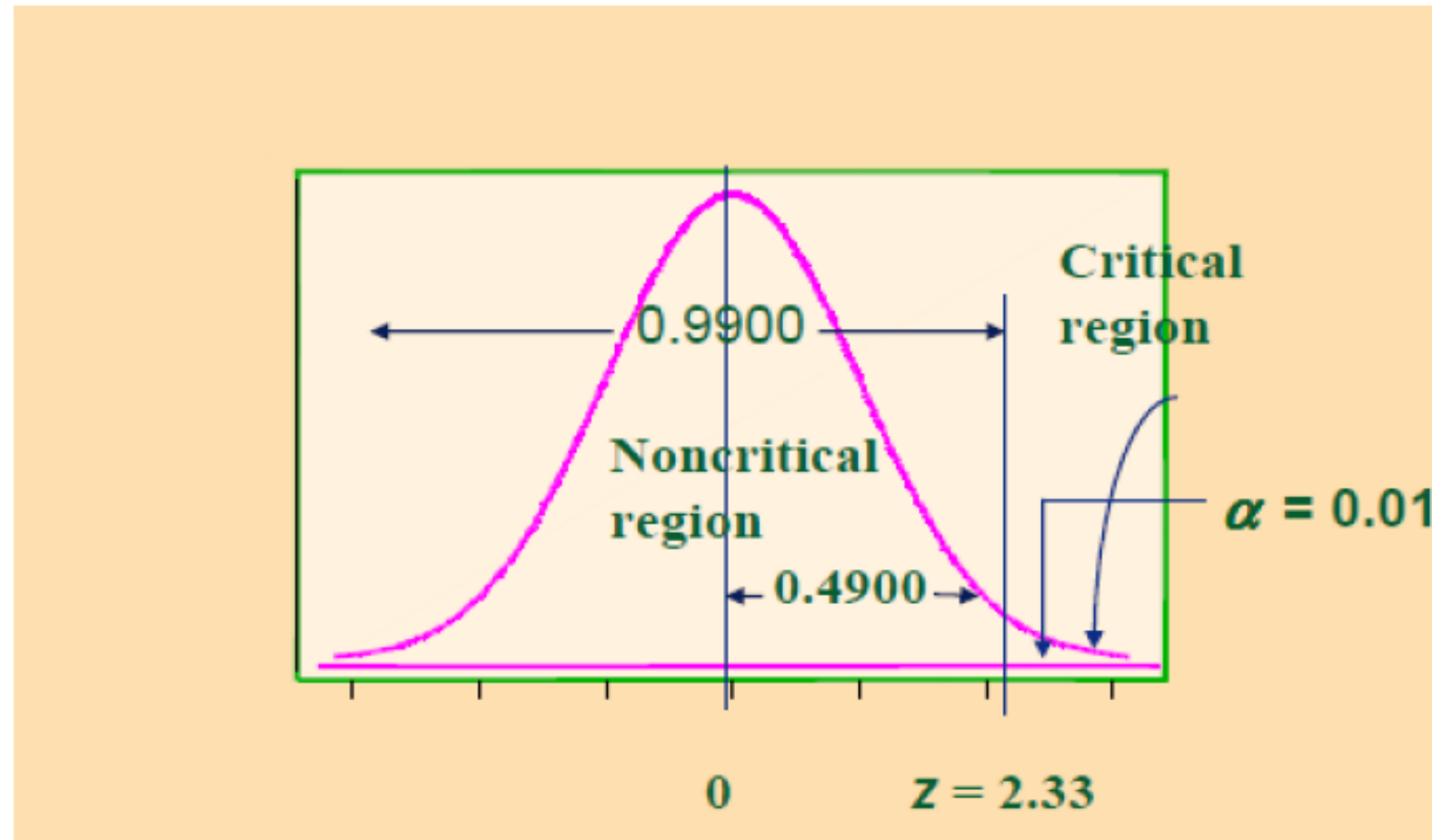
The **critical** or **rejection region** is the range of values of the calculated value that indicates that there is a significant difference and that the null hypothesis should be rejected.

Noncritical/Nonrejection Region:

- The region complementary to the critical region is called the **noncritical** or **nonrejection region**.
- The critical value separates the critical region from the noncritical region.

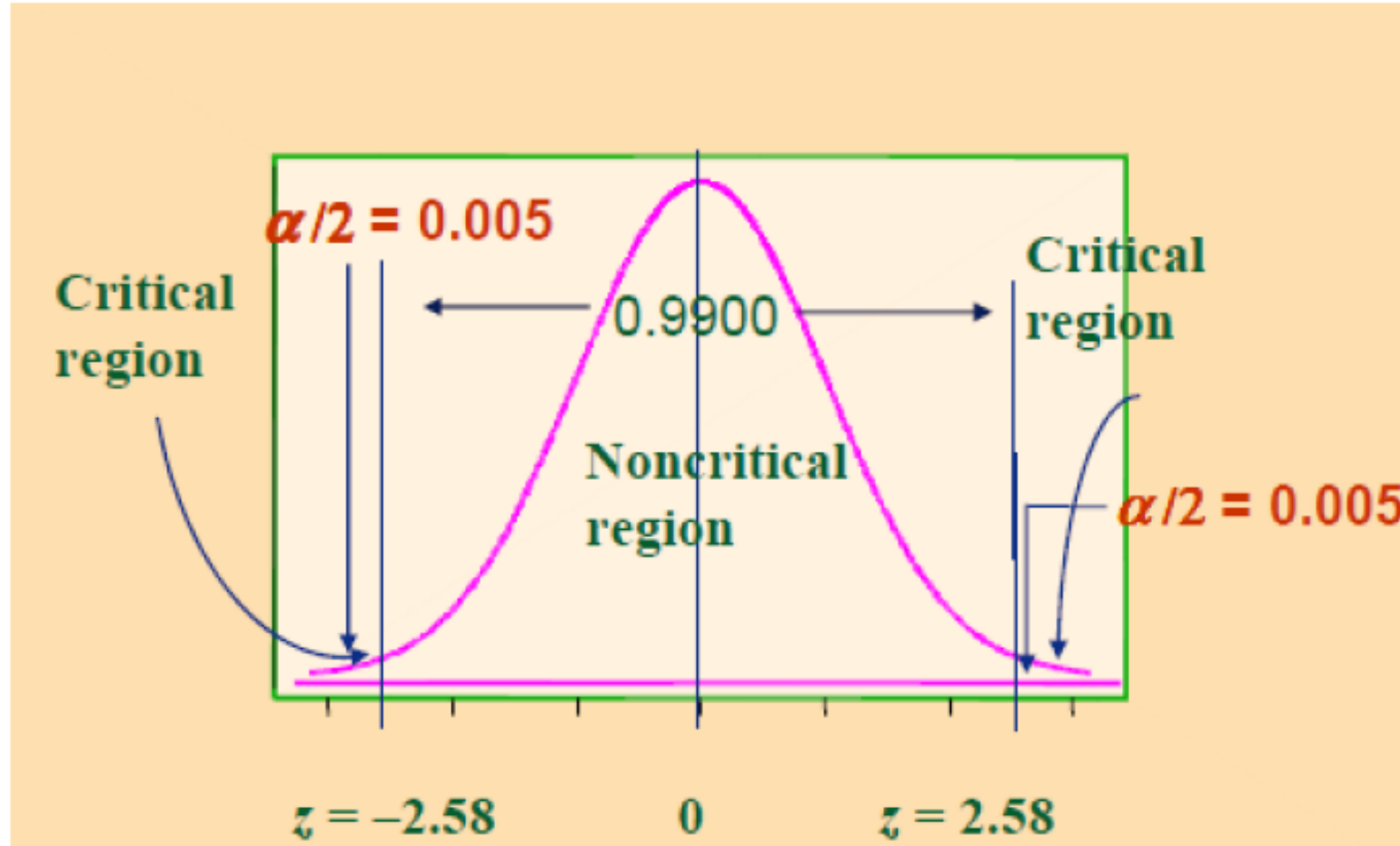
A one-tailed test (right or left) indicates that the null hypothesis should be rejected when the calculated value is in the critical region.

Critical value for $\alpha = 0.01$ (Right-Tailed Test):



In a two-tailed test, the null hypothesis should be rejected when the calculated value is in either of the two critical regions.

Critical value for $\alpha = 0.01$ (Two-Tailed Test):



Large Sample:

- If the sample size n is greater than or equal to 30 ($n \geq 30$), the sample is called a **Large Sample**.
- The z -test is a statistical test for the mean of a population. It can be used for large sample or when the population is normally distributed and σ is known.